STEM Fusion

Introduction to Quantum Computing

Applications & Impact

Quantum Algorithms · Hardware & Industry · Real-world Applications · Future Impact

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Innovation Hall 129

Schedule

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Homeroom

10:00 – 10:15

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G1 — Session Block

10:15 – 11:35

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Lunch

11:40 – 12:35

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G2 — Session Block

12:40 – 2:00

Student Expectations

NO FOOD	NO DRINKS	Be Curious
Respect Others	Ask Questions	Try & Fail
No Phones	Stay Focused	Have Fun!

Quantum Algorithms

The programs that make quantum computers powerful

Grover's Algorithm — The Quantum Spotlight 🔦

Grover's algorithm finds one specific item in an unsorted list dramatically faster than any classical approach.

The Problem — Searching Without a Map

Imagine 1 million unlabelled boxes. One contains a prize. You can only check one at a time.

Classical computer:
Check box by box, one at a time.
• Average: 500,000 checks
• Worst case: 1,000,000 checks

No shortcut exists — you simply have to get lucky or be extremely thorough.

For real applications like screening billions of drug candidates, or searching enormous databases, this is impossibly slow.

Grover's Solution — The Quantum Spotlight

Grover's uses superposition and interference:

1️⃣ Open ALL boxes at once. Superposition lets us consider every possibility simultaneously

2️⃣ Mark the prize box. An oracle subtly tags the correct answer

3️⃣ Amplify the marked box. Interference makes the correct answer's probability grow with each step

4️⃣ Measure — get the prize. After ~1,000 steps, the correct box has near-100% probability

Result: ~1,000 steps instead of ~500,000
That is a 500× speedup — proven optimal.

Shor's Algorithm — The Code Breaker 🔐

🔐 The Problem — RSA Encryption

The encryption protecting your bank account, messages, and government data relies on one fact:

Multiplying two huge prime numbers together is easy.
13 × 17 = 221 ← easy!

But finding which two numbers were multiplied — working backwards — is impossibly hard for classical computers when numbers are large enough.

A 2048-digit encryption key would take the world's fastest classical supercomputer longer than the age of the universe to crack.

All RSA encryption relies entirely on this remaining impossible.

⚛️ Shor's Quantum Solution

Shor's algorithm converts the factoring problem into a pattern-finding problem — and quantum computers find patterns exponentially faster using superposition.

For a 2048-digit number:
• Classical: longer than the age of the universe
• Quantum (when built): hours or days

Current status:
We need roughly 4 million physical qubits for a full attack on RSA-2048. We currently have ~1,000.

But the gap is closing fast. This is why governments worldwide are urgently deploying quantum-safe encryption NOW. The US published new quantum-safe standards in 2024.

Which Problems Quantum Algorithms Actually Solve 🎯

🔍 Searching — Grover's Algorithm

Finding one item in a large unsorted collection.

Classical: check one at a time
Quantum: explore all at once via superposition
Real uses: drug candidate screening, database search, financial optimisation
Speedup: 500× for 1 million items. 1,000× for 1 trillion items.

🔓 Factoring — Shor's Algorithm

Breaking apart large numbers into their prime components.

Classical: impossible for large numbers
Quantum: exponentially faster via quantum Fourier transform
Real uses: breaking RSA encryption, cryptanalysis
Why it matters: threatens all current internet encryption.

🧬 Simulation — Quantum Chemistry

Simulating how molecules interact at the atomic level.

Classical: approximation only — gets exponentially wrong
Quantum: exact simulation — molecules ARE quantum systems
Real uses: drug discovery, materials science, clean energy
This may be the most important quantum application of all.

🗺️ Optimisation — QAOA

Finding the best solution from an enormous number of options.

Classical: must guess or check systematically
Quantum: explores solution space using superposition
Real uses: delivery routing, airline scheduling, portfolio management, traffic flow

Hardware & Industry

The companies and countries racing to build quantum computers

The Major Players 🏆

🔵 IBM Quantum
Most Accessible

Largest quantum partner network — 450+ companies worldwide.

Free cloud access via IBM Quantum Lab — anyone can run real quantum experiments from a laptop.

Over 1,000 qubit processors available today.

Goal: 100,000 qubit system by 2033.

Uses superconducting qubits cooled near absolute zero.

🔴 Google Quantum AI
Quantum Supremacy Claim

In 2019, claimed their Sycamore processor solved a task no classical computer could match.

Willow chip (2024): 105 qubits, first to cross a critical error-correction threshold — a major milestone.

Heavy investment in combining quantum computing with artificial intelligence.

Uses superconducting qubits.

🟣 IonQ
Highest Accuracy

Uses trapped ions — individual charged atoms held in place by lasers.

Highest accuracy per operation of any major platform today.

First publicly traded quantum computing company (NYSE: IONQ).

Partners with Amazon Web Services and Microsoft Azure.

Slower than superconducting — but far more precise per gate.

Where We Are Now — The NISQ Era 🗺️

Today's quantum computers are useful but imperfect.
The industry calls this the NISQ era — Noisy Intermediate-Scale Quantum.

📍 Where We Are Today

- 100 – 1,000 physical qubits available
- Errors occur in ~1 in 100–1,000 operations
- Too noisy for most real commercial applications yet
- Actively demonstrating quantum principles and small proof-of-concept experiments
- IBM, Google, Amazon, Microsoft, IonQ all offer cloud access today
- You can run real quantum experiments from your laptop right now — for free

🎯 Where We're Going

- Millions of physical qubits needed for commercial applications
- Each protected 'logical' qubit requires ~1,000 physical qubits
- Error rates must drop 10–100× below today's levels
- Google's Willow chip (2024) first crossed a critical error threshold
- Estimated timeline: 10–15 years to fully fault-tolerant systems
- This unlocks Shor's algorithm, drug discovery, large-scale optimisation

🌍 The Global Race

- USA: $1.2B National Quantum Initiative Act
- EU: €1B Quantum Flagship programme
- China: estimated $15B+ investment, national strategic priority
- UK, Australia, Japan, South Korea all have major programmes
- Export controls on quantum hardware introduced in 2023
- Treated as a national security issue — not just a technology one

Real-World Applications

From saving lives to securing the internet

Six Industries Quantum Will Transform 🌍

💊 Medicine

Simulating drug-protein interactions exactly.
Could unlock cures for Alzheimer's, cancer, antibiotic resistance.
Pfizer, Roche, AstraZeneca all have active quantum programs.

🔐 Cybersecurity

Quantum threatens RSA — but also creates unbreakable quantum encryption.
NIST published quantum-safe standards in 2024.
Every bank and government must migrate NOW.

💰 Finance

Portfolio optimisation, risk simulation, fraud detection.
Goldman Sachs, JPMorgan Chase, HSBC all have quantum teams.
Faster Monte Carlo simulations for pricing complex products.

🌱 Climate & Energy

Designing better solar panels, batteries, and superconductors.
Simulating nitrogen fixation — could save 1–2% of global energy.
Key to reaching net-zero climate targets.

✈️ Logistics

Optimising delivery routes, airline scheduling, traffic flow.
VW cut Lisbon bus congestion 40% using quantum (2019).
Airbus using quantum for aircraft loading optimisation.

🤖 AI & Machine Learning

Quantum speedups could accelerate AI model training.
Google has TensorFlow Quantum available now.
Quantum machine learning: early stage but very active research.

Quantum-Safe Encryption — Why Urgent RIGHT NOW 🔒

Quantum computers do not fully exist yet — but the security threat they pose is real and active TODAY. Here is why.

📦 Harvest Now, Decrypt Later

Spy agencies and adversaries are recording large amounts of encrypted internet traffic today and storing it.

When large-scale quantum computers arrive in 10–20 years, they plan to decrypt all of it using Shor's algorithm.

Data sent today — medical records, classified documents, financial transactions — may be readable in the future.

This is a confirmed, documented threat. Not speculation.

🛡️ Post-Quantum Cryptography

In 2024, NIST published new encryption standards that quantum computers cannot break.

Based on mathematical problems even quantum computers cannot solve — like finding short paths in complex high-dimensional spaces.

Governments and companies worldwide must migrate all systems to these new standards before quantum computers arrive.

The migration itself takes years — which is exactly why it must start now, not later.

⏰ The Migration Timeline

2024: New quantum-safe standards published by NIST

2028–2030: US government agencies must migrate all systems

2035: Legacy RSA and ECC encryption officially retired

This affects every bank, every hospital, every government, every website using HTTPS.

Your generation will be the engineers and policymakers executing this global transition.

Future Impact

Technology · Society · Your world — what changes and when

Three Eras of Quantum Computing 🔭

Quantum computing is developing in three distinct phases — we are in the first one right now.

📍 NOW — The NISQ Era
2020 - 2030 (approx.)

- 100–1,000 noisy physical qubits
- Proofs-of-concept and small experiments
- Quantum Key Distribution (QKD) networks being built
- Cloud quantum access for researchers and companies
- First quantum advantages over classical demonstrated for specific tasks
- You can run a real quantum circuit on IBM hardware today — for free

🎯 MEDIUM TERM
2028 - 2035 (approx.)

- Early fault-tolerant quantum computers
- First real drug and materials simulation advantages
- Post-quantum cryptography fully deployed worldwide
- Quantum internet prototypes connecting cities
- Meaningful quantum advantages in financial modelling
- Breaking legacy RSA encryption becomes feasible
- Quantum computing moves from research to industry tool

🚀 LONG TERM
2035 and beyond

- Large-scale fault-tolerant systems
- Shor's algorithm breaks RSA-2048 at scale
- Quantum internet connecting continents
- Room-temperature quantum computing? (Speculative)
- Quantum AI: machine learning at quantum scale
- Simulating fundamental physics directly
- New materials and medicines impossible to discover classically
- The beginning of a post-quantum world

How Quantum Will Change Society 🌐

🔒 Global Security

Quantum threatens all current encryption. The 'harvest now, decrypt later' strategy means data stolen today is at risk in 10–20 years.

The US, EU, and allies are racing to deploy quantum-safe encryption before adversaries build large quantum systems.

2024: NIST standards published
2030: Government systems must migrate
2035: Legacy encryption retired

This is already shaping foreign policy and military strategy worldwide.

💊 Health & Medicine

Quantum simulation could unlock treatments for diseases with no current cure:

- Alzheimer's (50M+ affected)
- Antibiotic resistance (10M deaths/year by 2050 if unchecked)
- Many cancers
- Climate-related disease

Timeline: meaningful drug discovery advantages expected within 10–20 years as fault-tolerant systems emerge.

This could be one of the most significant impacts on human wellbeing in history.

⚖️ Ethics & Global Equity

Every major technology has widened inequality before eventually spreading. Quantum may follow the same pattern:

- Wealthy nations and companies build quantum first
- Strategic and economic advantages concentrate initially
- Smaller nations face new technological dependencies

Key questions your generation will answer:

- Should quantum capabilities be shared?
- Should quantum weapons be regulated like nuclear?
- Who governs the quantum internet?

There are no settled answers yet.